Computer Science - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following sets of bit strings CANNOT be described with a regular expression?
   (a) All bit strings whose number of zeros is a multiple of five
   (b) All bit strings starting with a zero and ending with a one
   (c) All bit strings with an even number of zeros
   (d) All bit strings with more ones than zeros

2. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
   (a) The wait is expected to be short.
   (b) A busy-wait loop is easier to code than an interrupt handler.
   (c) There is no other work for the processor to do.
   (d) The program executes on a time-sharing system.

3. Of the following, which gives the best upper bound for the value of f(N) where f is a solution to the recurrence f(2N + 1) = f(2N) = f(N) + log N for N >= 1, with f(1) = 0?
   (a) O(log N)
   (b) O(N log N)
   (c) O(log N) + 0(1)
   (d) O((log N)^2)

4. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?
   (a) To translate Web addresses to host names
   (b) To determine the IP address of a given host name
   (c) To determine the hardware address of a given host name
   (d) To determine the hardware address of a given IP address

5. Which of the following is true of interrupts?
   (a) They are generated when memory cycles are "stolen".
   (b) They are used in place of data channels.
   (c) They can indicate completion of an I/O operation.
   (d) They cannot be generated by arithmetic operations.

True / False (10 marks)

Answer True or False

6. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is n^m.
7. True or False: Reference counting requires additional space overhead for each memory cell to maintain count of references.
8. True or False: The symbol table is a data structure in a compiler that manages information about variables and their attributes.
9. True or False: Finding a shortest cycle in an undirected graph G can be solved in polynomial time.
10. True or False: A heap area is usually not represented in a subroutine's activation record frame for a stack-based programming language.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find numbers divisible by m or n from a list of numbers using lambda function.
12. Write a function to check whether the given number is undulating or not.

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